

VANGIPURAM VIBHUVAN

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EDUCATION

SASTRA Deemed University, Thanjavur, India

2023–2027

B.Tech in Computer Science and Engineering

Minor Specialization: Robotics and Automation

CGPA: **8.85 / 10**

PROJECTS

TradeApp – AI-Powered Trading Platform

GitHub

Node.js, Express, MongoDB, React, Socket.io, Jest, k6, GitHub Actions, RAG

- Built a real-time MERN trading platform featuring live portfolio tracking and synchronized market updates.
- Implemented ACID-compliant MongoDB transactions to ensure exact data consistency during concurrent order executions.
- Achieved a high success rate during 200-user k6 spike tests, reducing Express APIs to 60ms latency.
- Integrated a RAG-powered AI analytics copilot; ensured backend stability through an automated Jest CI pipeline via GitHub Actions.

Connectly – WebRTC Meet & Chat Application

GitHub

Node.js, Express, MongoDB, React, WebRTC, Socket.io

- Built a MERN communication platform using WebRTC for real-time peer-to-peer video and audio calling.
- Created dynamic room-based session management and live messaging using bidirectional Socket.io connections.
- Established direct P2P media connections, successfully bypassing server routing to minimize communication latency.
- Improved overall backend performance by offloading high-bandwidth streaming, significantly reducing central server load.

Astra RAG – Containerized Agentic AI Platform

GitHub

Node.js, Express, MongoDB Vector Search, React, LangGraph, Docker, Ollama

- Developed an Agentic AI platform integrating LLM chat, internet search, and RAG-based document retrieval.
- Designed LangGraph workflows with HuggingFace embeddings and MongoDB Atlas Vector Search.
- Implemented intelligent tool routing for seamless query handling across multiple AI capabilities.
- Containerized the MERN application using Docker with local Llama 3 inference via Ollama.

SKILLS

Languages: C, C++, Java (Core), JavaScript (ES6+), Python, SQL

Tech Stack: React, Node.js, Express.js, Docker

Databases: MongoDB (NoSQL), MySQL (Relational)

AI: LLM Integration, RAG, Prompt Engineering, Vector Databases LangChain, LangGraph

Tools: Docker, Git, GitHub, GitHub Actions, Jest, k6

Core CS: DSA, OOP, DBMS, OS, Computer Networks

DSA

- Solved **300+ LeetCode problems (C++)** covering Dynamic Programming, Graphs, Trees, and Greedy algorithms.
- Profile: leetcode.com/u/Mittu_2006